

Q6) FAIRGAME: a Framework for AI Agents Bias Recognition using Game Theory

1. What is the role of Game theory in AI agent collaboration?

Game theory helps model and analyse collaboration between AI agents by defining their possible strategic moves and rewards. The researchers predict how rational agents should behave and compare that with how the LLM based agents cooperate, compete and coordinate helping identify the unexpected behavior.

2. What is the importance of game theory?

It is important because it helps us study how AI agents make decisions when they interact with each other by providing a mathematical framework. It predicts whether it will cooperate or compete or and allows researchers to identify unexpected behaviors when they do not behave according to theory.

3. What are the two classical Game theories presented in the paper? And how it can be used for the 5G applications?

Prisoner's Dilemma and Battle of Sexes are the 2 game theories used in the paper. Prisoner's Dilemma is mainly about cooperation vs. selfish behavior, while battle of sexes is about coordination when 2 sides are preferring different outcomes. In 5G these ideas can be used in things like resource allocation, and coordination between network agents or base stations.

4. Enlist selected LLMs models presented in a paper.

- Llama 3.1 405B
- Mistral Large
- GPT-4
- Claude 3.5 Sonnet

5. [slides for fairgame](#)

Q6) A semi-independent policies training method with shared representation for heterogeneous multi-agents reinforcement learning

1. How you can describe POMDP mathematically ?

POMDP is a mathematical model for making decisions when you cannot see the true state of the world. Written as : $(S, A, T, R, \Omega, O, \gamma)$.

S: possible states of the world

A: actions the agent can take

T : chance that action a moves the world from state s to s'

R: reward for taking that action

Ω : possible observations

O: chance of seeing observation o after the action

γ : how much future rewards count

2. What are the application of Hard and soft parameter sharing?

In Hard parameter sharing tasks use lower layers of a neural network then have separate upper layer for their own decisions. In multi-agent method, different types of agents share a common base while each type has its own policy helping them share useful experiences and reducing the amount of training need.

On the other hand, in soft parameter sharing each task has its own models and parameter , giving the tasks more freedom to specialise.

3. Explain and decode equation no 1.

$$\mathcal{L}_e(\theta_e) = \mathbb{E}_{(o, a, r, o') \sim \mathcal{D}} \left[\sum_i \|f_o(z_{a_i}, o_i, \mathbf{a}_{-i}) - o'_i\|_2^2 + \lambda_e \sum_i (f_r(z_{a_i}, o_i, \mathbf{a}_{-i}) - r)^2 \right] \quad (1)$$

Eq 1 measures prediction error. The model predicts each agent's next observation and reward, then compares both with what actually happened. It adds the errors across agents and training examples. λ_e controls how much reward error counts.

Training aims to make this total error as small as possible.

4. Explain intuition behind figure no 4 presented in the paper.

It shows a learning loop: agents act, environment responds and the agent receives usual reward. One reward encourages agent to explore unfamiliar situation and the other helps model tell different agent types apart. thick black lines illustrate data flow, thin black lines illustrate rewards and red lines illustrate the loss to generate the intrinsic rewards. These rewards guide what agents learn next.

5. [slides](#)